The following program models vehicles using classes, objects, and inheritance structures. It demonstrates Object Oriented Programming (OOP) principles, such as inheritance, polymorphism, and encapsulation. The relationship between the Vehicle base class and the Car, Truck, and Motorcycle subclasses demonstrates inheritance. These subclasses, representing different object types, demonstrate polymorphism (McGrath, 2021). The displayInfo() and compare() functions in the Vehicle base class, being used to operate on each subclass, are an example of encapsulation. The Scope Resolution Operator (::) is critical to accessing the functions of the parent Vehicle class (PHP Group, 2025). As for the specific functions, the comments in the code explain their intent. The program is organized in accordance with its flow. It begins with the Vehicle base class, branches out into the Car, Motorcycle, and Truck subclasses, and ends with the HTML form as well as the PHP code to handle its input.

Overall, the program's organization should make it easier for developers to scale their management system. They could add new Vehicle types, such as jet skis or tractors, for example. The new subclasses would extend the Vehicle class, enhancing code reusability and memory efficiency. Adherence to the core OOP principles will help in this endeavor.

Code screenshots:

```php
<?php
//BASE CLASS
class Vehicle {
    public $brand; // Vehicle characteristic variables
    public $model;
    public $year;
    public $price;
    public static $vehicleCount = 0; // Tracks number of created vehicles
    public static $vehicleList = []; // Stores all created vehicle objects

    // Constructor: Instantiates vehicle with common vehicle characteristics
    public function __construct($brand, $model, $year, $price) {
        $this->brand = $brand;
        $this->model = $model;
        $this->year = $year;
        $this->price = $price;
        self::$vehicleCount++;
        self::$vehicleList[] = $this; // Adds to the static list
    }

    // displayInfo() function: Displays common vehicle details
    // Intent: Output attributes shared by all vehicle types
    public function displayInfo() {
        echo "<strong>Vehicle Details:</strong><br>";
        echo "Brand: {$this->brand}<br>";
        echo "Model: {$this->model}<br>";
        echo "Year: {$this->year}<br>";
        echo "Price: \${$this->price}<br>";
    }

    // compare() function: Compare two vehicles based on criterion
    // Intent: Provide simple sorting or comparison by year or price
    public static function compare($v1, $v2, $criterion) {
        if ($criterion == 'price') {
            return $v1->price <=> $v2->price;
        } elseif ($criterion == 'year') {
            return $v1->year <=> $v2->year;
        }
        return 0;
    }
}
```

```php
// SUBCLASS: Car
class Car extends Vehicle {
    public $numberOfDoors;

    // Constructor: Adds unique property for number of doors
    public function __construct($brand, $model, $year, $price, $numberOfDoors) {
        parent::__construct($brand, $model, $year, $price);
        $this->numberOfDoors = $numberOfDoors;
    }

    // displayInfo(): Adds unique car details
    // Intent: Show both shared and car-specific info
    public function displayInfo() {
        parent::displayInfo();
        echo "Number of Doors: {$this->numberOfDoors}<br><hr>";
    }
}

// SUBCLASS: Truck
class Truck extends Vehicle {
    public $cargoCapacity;

    // Constructor: Adds unique property for cargo capacity
    public function __construct($brand, $model, $year, $price, $cargoCapacity) {
        parent::__construct($brand, $model, $year, $price);
        $this->cargoCapacity = $cargoCapacity;
    }

    // displayInfo(): Adds truck-specific info
    // Intent: Extend base display to include truck details
    public function displayInfo() {
        parent::displayInfo();
        echo "Cargo Capacity: {$this->cargoCapacity} tons<br><hr>";
    }
}

// SUBCLASS: Motorcycle
class Motorcycle extends Vehicle {
    public $handlebarType;

    // Constructor: Adds unique property for handlebar type
    public function __construct($brand, $model, $year, $price, $handlebarType) {
        parent::__construct($brand, $model, $year, $price);
        $this->handlebarType = $handlebarType;
    }

    // displayInfo(): Adds motorcycle-specific info
    // Intent: Extend base display to include motorcycle details
    public function displayInfo() {
        parent::displayInfo();
        echo "Handlebar Type: {$this->handlebarType}<br><hr>";
    }
}
```

```php
// HTML form submission handling
if ($_SERVER["REQUEST_METHOD"] == "POST") {

    // if statement to handle adding a new vehicle
    if (isset($_POST['add'])) {
        $type = $_POST['type'];
        $brand = $_POST['brand'];
        $model = $_POST['model'];
        $year = $_POST['year'];
        $price = $_POST['price'];
        $unique = $_POST['unique'];

        // Switch case to create the correct subclass based on the selected type
        switch ($type) {
            case 'car':
                new Car($brand, $model, $year, $price, $unique);
                break;
            case 'truck':
                new Truck($brand, $model, $year, $price, $unique);
                break;
            case 'motorcycle':
                new Motorcycle($brand, $model, $year, $price, $unique);
                break;
        }

        echo "<h3>Vehicle Added Successfully!</h3>";
        echo "<p><strong>Total Vehicles Created:</strong> " . Vehicle::$vehicleCount . "</p>";
        echo '<form method="post">
                <input type="hidden" name="display" value="1">
                <input type="submit" value="Display All Vehicles">
              </form>';
    }

    // When displaying all vehicles in the management system
    if (isset($_POST['display'])) {
        echo "<h3>All Vehicles in the System:</h3>";
        if (empty(Vehicle::$vehicleList)) {
            echo "<p>No vehicles have been added yet.</p>";
        } else {
            foreach (Vehicle::$vehicleList as $v) {
                $v->displayInfo();
            }
        }
        echo "<hr><a href=''>← Go Back to Add Another Vehicle</a>";
        exit; // stop further HTML from showing
    }
}
?>
```

```html
<!-- HTML Form for the user to add new vehicles -->
<h2>Add a New Vehicle</h2>
<form method="post">
    <label>Type:</label><br>
    <select name="type" required> <!-- Select between vehicle type -->
        <option value="car">Car</option>
        <option value="truck">Truck</option>
        <option value="motorcycle">Motorcycle</option>
    </select><br><br>

    <!-- User-friendly fields for the vehicle attributes -->
    <label>Brand:</label><br>
    <input type="text" name="brand" required><br><br>

    <label>Model:</label><br>
    <input type="text" name="model" required><br><br>

    <label>Year:</label><br>
    <input type="number" name="year" required><br><br>

    <label>Price ($):</label><br>
    <input type="number" name="price" step="0.01" required><br><br>

    <label>Unique Attribute (Doors / Cargo Capacity / Handlebar Type):</label><br>
    <input type="text" name="unique" required><br><br>

    <input type="submit" name="add" value="Add Vehicle">
</form>
```

Code text:

```
<?php

//BASE CLASS

class Vehicle {

    public $brand; // Vehicle characteristic variables

    public $model;

    public $year;

    public $price;
```

```
public static $vehicleCount = 0; // Tracks number of created vehicles
public static $vehicleList = []; // Stores all created vehicle objects

// Constructor: Instantiates vehicle with common vehicle characteristics
public function __construct($brand, $model, $year, $price) {
    $this->brand = $brand;
    $this->model = $model;
    $this->year = $year;
    $this->price = $price;
    self::$vehicleCount++;
    self::$vehicleList[] = $this; // Adds to the static list
}

// displayInfo() function: Displays common vehicle details
// Intent: Output attributes shared by all vehicle types
public function displayInfo() {
    echo "<strong>Vehicle Details:</strong><br>";
    echo "Brand: {$this->brand}<br>";
    echo "Model: {$this->model}<br>";
    echo "Year: {$this->year}<br>";
    echo "Price: \${$this->price}<br>";
}
```

```
    // compare() function: Compare two vehicles based on criterion
    // Intent: Provide simple sorting or comparison by year or price
    public static function compare($v1, $v2, $criterion) {
        if ($criterion == 'price') {
            return $v1->price <=> $v2->price;
        } elseif ($criterion == 'year') {
            return $v1->year <=> $v2->year;
        }
        return 0;
    }
}

// SUBCLASS: Car
class Car extends Vehicle {
    public $numberOfDoors;

    // Constructor: Adds unique property for number of doors
    public function __construct($brand, $model, $year, $price, $numberOfDoors) {
        parent::__construct($brand, $model, $year, $price);
        $this->numberOfDoors = $numberOfDoors;
    }

    // displayInfo(): Adds unique car details
```

```
    // Intent: Show both shared and car-specific info
    public function displayInfo() {
        parent::displayInfo();
        echo "Number of Doors: {$this->numberOfDoors}<br><hr>";
    }
}

// SUBCLASS: Truck
class Truck extends Vehicle {
    public $cargoCapacity;

    // Constructor: Adds unique property for cargo capacity
    public function __construct($brand, $model, $year, $price, $cargoCapacity) {
        parent::__construct($brand, $model, $year, $price);
        $this->cargoCapacity = $cargoCapacity;
    }

    // displayInfo(): Adds truck-specific info
    // Intent: Extend base display to include truck details
    public function displayInfo() {
        parent::displayInfo();
        echo "Cargo Capacity: {$this->cargoCapacity} tons<br><hr>";
    }
```

```
}

// SUBCLASS: Motorcycle
class Motorcycle extends Vehicle {
    public $handlebarType;

    // Constructor: Adds unique property for handlebar type
    public function __construct($brand, $model, $year, $price, $handlebarType) {
        parent::__construct($brand, $model, $year, $price);
        $this->handlebarType = $handlebarType;
    }

    // displayInfo(): Adds motorcycle-specific info
    // Intent: Extend base display to include motorcycle details
    public function displayInfo() {
        parent::displayInfo();
        echo "Handlebar Type: {$this->handlebarType}<br><hr>";
    }
}

// HTML form submission handling
if ($_SERVER["REQUEST_METHOD"] == "POST") {
```

```
// if statement to handle adding a new vehicle
if (isset($_POST['add'])) {
    $type = $_POST['type'];
    $brand = $_POST['brand'];
    $model = $_POST['model'];
    $year = $_POST['year'];
    $price = $_POST['price'];
    $unique = $_POST['unique'];

    // Switch case to create the correct subclass based on the selected type
    switch ($type) {
        case 'car':
            new Car($brand, $model, $year, $price, $unique);
            break;
        case 'truck':
            new Truck($brand, $model, $year, $price, $unique);
            break;
        case 'motorcycle':
            new Motorcycle($brand, $model, $year, $price, $unique);
            break;
    }

    echo "<h3>Vehicle Added Successfully!</h3>";
```

```
    echo "<p><strong>Total Vehicles Created:</strong> " . Vehicle::$vehicleCount . "</p>";
    echo '<form method="post">
            <input type="hidden" name="display" value="1">
            <input type="submit" value="Display All Vehicles">
          </form>';
  }

  // When displaying all vehicles in the management system
  if (isset($_POST['display'])) {
    echo "<h3>All Vehicles in the System:</h3>";
    if (empty(Vehicle::$vehicleList)) {
      echo "<p>No vehicles have been added yet.</p>";
    } else {
      foreach (Vehicle::$vehicleList as $v) {
        $v->displayInfo();
      }
    }
    echo "<hr><a href="">← Go Back to Add Another Vehicle</a>";
    exit; // stop further HTML from showing
  }
}
?>
```

```
<!-- HTML Form for the user to add new vehicles -->
<h2>Add a New Vehicle</h2>
<form method="post">
    <label>Type:</label><br>
    <select name="type" required> <!-- Select between vehicle type -->
        <option value="car">Car</option>
        <option value="truck">Truck</option>
        <option value="motorcycle">Motorcycle</option>
    </select><br><br>

            <!-- User-friendly fields for the vehicle attributes -->
    <label>Brand:</label><br>
    <input type="text" name="brand" required><br><br>

    <label>Model:</label><br>
    <input type="text" name="model" required><br><br>

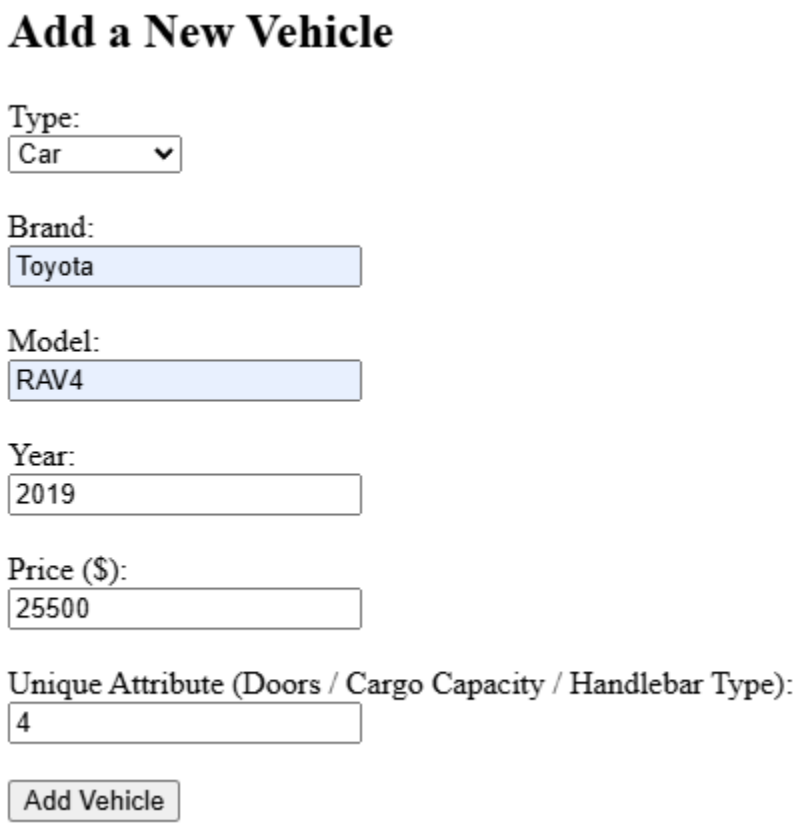
    <label>Year:</label><br>
    <input type="number" name="year" required><br><br>

    <label>Price ($):</label><br>
    <input type="number" name="price" step="0.01" required><br><br>
```

```
    <label>Unique Attribute (Doors / Cargo Capacity / Handlebar Type):</label><br>
    <input type="text" name="unique" required><br><br>

    <input type="submit" name="add" value="Add Vehicle">
</form>
```

Output example:



References:

McGrath, M. (2021). *PHP in easy steps (4th ed.): Updated for PHP 8*. In Easy Steps Limited.

PHP Group. (2025). Classes and Objects. *PHP Manual*. Licensed Under Creative Commons Attribution 3.0. https://www.php.net/manual/en/language.oop5.php